CHRISTIAN KEOUGH

Gainesville, FL | christiankeough1@gmail.com | Portfolio | LinkedIn

EDUCATION

Bachelor of Science in Mechanical Engineering **May 2028**
Minor in Computer Science GPA: 3.86
University of Florida, Gainesville, FL
Relevant Coursework: Statics, Computer Aided Graphics and Design, Thermodynamics, Physics with Calc 2, MATLAB, Materials

Associate in Engineering **April 2024**
Indian River State College, Fort Pierce, FL GPA: 3.93

Honors: Valedictorian, Class of 800 students (2024); Dean's List (2023-Present); AICE Diploma with Merit

PROFESSIONAL EXPERIENCE

Florida Advanced Manufacturing and Systems Integration Laboratory
Undergraduate Lab Assistant March 2026 – Present
- Integrating Epson 6-axis robot, motion controller, and vision camera to allow conformal printing with prosthetics and implants.
- Built a SolidWorks-Epson pipeline, converting CAD curves into 3-axis robot paths with SPEL+, for prosthetic printing.
- Programmed G-code + Hansen Solubility interface to reduce RED calc time by 95% for DIW VIPS multi-layer lattice printing.
- Recorded phase separation in real time via optical microscopy across 30 experiments to guide future VIPS design.

Fastenal, Onsite at Sandvik Mining
Customer Site Warehouse Support September 2026 – Present
- Process 2 weekly inbound shipments of 100+ parts each, logging them into inventory system to keep stock accurate.
- Fulfill orders of up to 5,000 parts for 8+ production areas in 4 warehouses, restocking vending machines via Zebra scanners.

PGA Homes and Roofing
Roofing Assistant Summer 2024 & Summer 2025
- Installed and reinforced tile roof systems using bull tar, membrane, polyurethane foam; improved durability of 20 roof sections.

INVOLVEMENT

Generational Relief in Prosthetics (GRiP)
Tendon-Driven Hand, Production Design, 3D Printing Teams Spring 2026 – Present
- Designed child bike pedal prosthetic in SolidWorks from 15+ ImageJ-measured dimensions, performed TPU/PLA 3D Printing.
- Modeling palm and structural components in SolidWorks for a tendon-driven prosthetic hand with haptic feedback.

Design Build Fly (DBF)
Structures Subteam Member Fall 2025
- Modeled aileron skin geometry in SolidWorks; iterated ~5 designs, collaborated with wing subteam for fit + manufacturability.

FIRST Robotics Competition (FRC), VEX Robotics Program
Mechanical Lead / Vice President Fall 2020 – Spring 2022
- Coordinated design and assembly of VEX robots, integrated sensors, drivetrain across 2 robot iterations (autonomous + driver).

PROJECTS

Air Engine Fabrication & Food Mixer Prototype, Design and Manufacturing Lab Fall 2026
- Drafting engineering drawings and machining 16 air engine parts on turret lathes and mills to ±0.005 in tolerance.
- Designed 24-part food mixer prototype, sourcing 12 McMaster-Carr parts and sizing fasteners via tap/drill charts.

Canine Pelvic Brace with Hip Socket CAD Design Spring 2026
- Modeled CAD canine pelvic brace from CT scans via 3D splines, anatomical planes; captured ~25 anatomical references.
- Integrated hip socket assistive device with joint articulation and structural analysis using Abaqus; ran 3 simulation validations.

Hansen Solubility Material Selection Interface Spring 2026 – Present
- Developed Python tool parsing CSV solvent data, analyzing ~30 solvents, and generating 3D/2D compatibility plots.
- Applied Hansen solubility interface for DIW VIPS solvent selection; screened 30 comparisons for most optimal pairing.

SKILLS & CERTIFICATIONS

SolidWorks (CSWA), Inventor (2022), Onshape, Epson SPEL+, Python, C++, MATLAB, Abaqus FEA, Lathe/Mill, 3D Printing